

Tiya Agarwal

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Computer Science undergraduate specializing in full-stack and AI/ML application development, with hands-on experience across **REST APIs**, **RAG pipelines**, and production deployment. Strong in system design, testing, and **200+ DSA problems solved**.

Education

Vellore Institute of Technology, Vellore

Aug 2023 – Jul 2027

B.Tech, Computer Science & Technology — CGPA: 8.62/10

Experience

Software Engineering Intern

May 2026 – Jul 2026

DeepFold Technologies, Gurugram

- Validated **REST API** schemas and request/response contracts across modules, catching defects pre-production.
- Debugged frontend/backend defects in **JavaScript** and **SQL**, fixing UI bugs and query errors for reliability.

Software & IT Solutions Intern

May 2025 – Jul 2025

Elegant Soft Solutions & Services, Agra

- Wrote **SQL** queries to validate migrated **ERP** records against source data during a large-scale **TallyPrime** migration project.
- Tested inventory and accounting workflows end-to-end, logging defects and running data-integrity checks ahead of go-live.

Projects

RootPilot | *Java 21, Spring Boot 4, PostgreSQL, Redis, RabbitMQ, React + TypeScript*

AIOps / Observability

- Architected **RootPilot**, an **AIOps** platform with **Spring Boot** and **React**, flagging anomalies via **30-point Z-score**.
- Persisted incidents to **PostgreSQL** with a **Redis** cache and fallback, exposed via **JWT-secured REST API**.
- Engineered a **Gemini-powered Copilot** endpoint for root-cause analysis, plus a **React Flow** dashboard.

SageVault | *Node.js, Express 5, MongoDB, React 19, LangChain, FAISS, Jest/CI*

Full-Stack / GenAI

- Built SageVault, a **RAG** platform querying PDFs/Word docs via **LangChain** and **FAISS** before **Gemini**.
- Designed a **JWT**-authenticated, rate-limited **REST API** with **fail-fast** startup validation, plus an analytics dashboard.
- Validated the API/pipeline via a **31-file Jest/Supertest suite** in **GitHub Actions CI**, plus an ADR log.

MockMate | *FastAPI, MongoDB Atlas, React, Gemini, Vertex AI, AWS S3*

Full-Stack / GenAI

- Built MockMate, a **FastAPI/React** platform where candidates answer **Gemini**-generated questions via **MediaRecorder**.
- Engineered a video pipeline uploading to **AWS S3**, invoking **Vertex AI Gemini** for transcription/scoring via modules.
- Delivered AI-scored feedback as structured JSON, covering **0–100** content scores and a **0–10** communication score.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, HTML5, CSS3, SQL, C++, C

Frameworks: React.js, Node.js, Express.js, Spring Boot, FastAPI, Flask

Databases & Messaging: MongoDB, PostgreSQL, SQLite, Redis, RabbitMQ

Cloud & Tools: AWS S3, Docker, Git, GitHub Actions CI, Railway, Vercel, REST API Design, JWT

Testing: Jest, Supertest, pytest, Postman

Core Concepts: Data Structures & Algorithms, System Design, OOP, DBMS, Operating Systems, Distributed Systems

Certifications

AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services, covering cloud architecture and security (2026)

Machine Learning — SmartBridge x Google for Developers, covering supervised learning and deployment workflows (2025)

Journey to Cloud: Envisioning Your Solution — IBM SkillsBuild, cloud solution design and migration strategy (2025)

Getting Started with Artificial Intelligence — IBM SkillsBuild, covering AI fundamentals and use-case design (2025)

Cyber Security — Eduversity, covering security principles, threat modeling, and mitigation practices (2023)

Achievements & Leadership

Solved **200+ DSA problems** on LeetCode across Easy, Medium, and Hard levels, strengthening algorithmic problem-solving.

Competed in **4+ hackathons**, rapidly prototyping full-stack and ML solutions end-to-end under tight time constraints.

Core Member, IEEE Computer Society — led AI/ML discussions and peer learning sessions for **50+ members** year-round.

Public Relations Head, VIT Dance Team — led event outreach, branding, and publicity for campus-wide performances.